

SMART INDIA HACKATHON 2025

Idea Submission

SETU

The allocation layer between the warning and the rescue

Problem Statement ID — [PS ID from portal]

Problem Statement Title — Real-Time Disaster Early-Warning & Resource Coordination Platform

Theme — Disaster Management

PS Category — Software

Team ID — [Team ID from portal]

Team Name (Registered on portal) — [Team Name]

IDEA TITLE

Proposed Solution — SETU: a district disaster-coordination platform

What it does

- ▶ Detection and dissemination are solved nationally (IMD, SACHET). Coordination is not — it runs on phone calls and paper.
- ▶ SETU takes citizen reports on 4 channels, deduplicates them into incidents, and solves the allocation problem optimally.
- ▶ One map-based DEOC dashboard: incoming reports and available resources on the same picture, with live dispatch lines.

How it addresses the problem

- ▶ 200 reports collapse into 28 real incidents — the DEOC stops dispatching against duplicate noise.
- ▶ Every incident is scored on 5 explainable terms, including a population equity term so relief does not follow phone density.

Innovation and uniqueness

- ▶ Every team demos a map with pins. SETU demos a solver — and proves it against the baseline, live on screen.
- ▶ Two different solvers because they are two different problems: Hungarian assignment for rescue (1-to-1), min-cost flow for shelters and relief supplies (capacitated, many-to-many).
- ▶ Operations research, not ML: deterministic, explainable and auditable — the correct choice for a life-safety system.
- ▶ Offline by design. The whole stack, including a 48 MB vector basemap, runs with the network unplugged.

14%

faster response*

+3more incidents
served***16**

districts covered

1,100 km

coastline

4

ingest channels

110 ms

solver cycle

42 ms

ingest latency

* measured on the running prototype, like-for-like over the incidents both strategies serve (28 incidents / 40 units, Ganjam cyclone scenario)

TECHNICAL APPROACH

Technologies used

Frontend React · Vite · MapLibre GL · Zustand · PWA (Workbox, IndexedDB)

Backend Python · FastAPI · SQLAlchemy (async) · WebSockets

Data PostgreSQL 16 + PostGIS · Redis (queue, pub/sub, cache)

Intelligence SciPy linear_sum_assignment (Hungarian) · NetworkX max-flow-min-cost · ST_ClusterDBSCAN

Maps / Infra Protomaps PMTiles (offline basemap) · OpenStreetMap · Docker Compose

Methodology — one pipeline, from a phone to a dispatched boat



- ▶ Debounced 2 s optimisation loop, per-district Redis lock, commitment locking so a committed unit is never reassigned (anti-thrash).
- ▶ Hard constraints (capability, availability) enforced twice — in SQL and re-asserted in the solver. Capacity is a preference, never a veto: a smaller boat is worse than a bigger one, never worse than nobody.
- ▶ Degradation is designed, not accidental: OSRM → haversine ETAs, live CAP → cached last-good, GPS → typed place name → PIN code, online → IndexedDB outbox.

FEASIBILITY AND VIABILITY

Feasibility — already built and measured

- ▶ Working prototype, not a concept: 6 services, one `docker compose up`, 36 automated tests passing.
- ▶ Full optimisation cycle 110–170 ms against 28 incidents and 40 units — three orders of magnitude inside the 1.5 s target.
- ▶ Seeded across 16 coastal districts: 381 shelters, 326 units, 73,664 seed rows, in 3 seconds.
- ▶ No new hardware. Runs on one laptop; a district needs a mid-range VM.

Challenges and risks

- ▶ No connectivity in the disaster zone — the exact moment the system is needed.
- ▶ Report flooding and panic-forwarding skewing triage.
- ▶ False or duplicated reports wasting scarce units.
- ▶ Adoption: an authority will not depend on a black box it cannot justify.
- ▶ Data gaps — no public NDRF/ODRAF asset register exists.

Strategies for overcoming them

Connectivity 4-rung degradation ladder: app → SMS → IVR → field unit. All four produce the same record; the solver never learns which.

Report flooding Corroboration is log-scaled and capped — 200 forwarded messages cannot outweigh one genuine report.

False reports Trust scoring gates a visible quarantine queue at 0.35. Low-trust reports are never auto-dispatched and never dropped.

Adoption Every score ships its 5-term breakdown; every commit and override is written to an append-only audit log.

Data gaps Real place names and coordinates; synthetic rosters labelled as such. Real registers swap in as data, not code.

IMPACT AND BENEFITS

Potential impact on the target audience

Citizens A return channel that works with no internet. A reference code, a tracked status, and the nearest shelter with free beds.

DEOC / Collector One operating picture instead of WhatsApp and a paper register. Every dispatch justified by a visible score.

NDRF / ODRAF Units sent where they do the most good, not where the loudest call came from. No unit double-tasked.

SDMA / State Shelter and relief shortfall surfaced as a live number — an escalation signal no existing system produces.

Benefits

- ▶ Social — the golden hours between warning and rescue are where deaths occur. Faster, fairer allocation converts directly into lives.
- ▶ Equity — report volume tracks phone density, not need. A population term stops relief concentrating in connected wards.
- ▶ Economic — no new hardware, open-source stack, no per-map licence.

Scale and sustainability

- ▶ Multi-tenant from day one — 10 tables keyed by district, one solver loop each. A new district is a seed file, not a rewrite.
- ▶ 16 districts across 3 states already running: Odisha, Andhra Pradesh, West Bengal — the full cyclone landfall corridor.
- ▶ Auditable by construction: an append-only log turns the event into a compensation and claims record afterwards.

RESEARCH AND REFERENCES

Standards and official sources

- NDMA SACHET — Common Alerting Protocol (CAP 1.2) public feed · sachet.ndma.gov.in
- India Meteorological Department — rainfall classes and wind (gale/storm) scales, used for the hazard outlook
- Sphere Handbook — Minimum Standards in Humanitarian Response: 15 L water and 0.5 kg ration per person per day
- OSDMA — Odisha multipurpose cyclone shelter programme, capacities and siting

Methods

- Kuhn (1955), The Hungarian method for the assignment problem — optimal 1-to-1 dispatch, $O(n^3)$
- Ester et al. (1996), DBSCAN — density-based clustering; applied via PostGIS ST_ClusterDBSCAN, eps 300 m
- Min-cost flow (Busacker–Gowen / network simplex) — capacitated shelter evacuation and relief distribution, via NetworkX

Data and technology

- OpenStreetMap (ODbL) via Protomaps PMTiles — offline vector basemap, 47.6 MB for the corridor
- PostGIS 3.4 — all spatial work in SQL; SciPy linear_sum_assignment; MapLibre GL JS
- Historical landfalls informing the corridor: 1999 Odisha Super Cyclone, Phailin 2013, Hudhud 2014, Fani 2019, Amphan 2020, Yaas 2021

Working prototype · github.com/Aditya365x/setu2.0 — 6 services, 36 passing tests, 16 districts seeded, runs offline via docker compose